

① Is AI leading to lay-offs by EC? } not be better highlighted
↓
wage growth

Kristóf András
Eszter Riesing

Automation, AI and Labor Market Outcomes in the European Union

② Data is limited

1. Introduction

Rapid advances in robotics and artificial intelligence (AI) are transforming the way work is organized. The integration of AI and robotics into the global economy is reshaping labor markets, altering the structure of employment, and challenging traditional economic models. The central debate in the existing literature concerns the extent to which automation displaces workers or complements human labour in ways that increase productivity and create new employment opportunities. A parallel body of research investigates why the consequences of these technologies differ across regions and countries, highlighting the role of economic complexity and structural economic resilience. This proposal integrates these strands by examining how robotics and AI influence labour-market outcomes and how economic complexity mitigates the potentially adverse effects of automation.

Recent studies highlight the dual impact of AI and robotics, showing increased productivity and the displacement of routine manual jobs (Acemoglu & Restrepo, 2020; Bharadwaj & Dvorkin, 2019). For instance, evidence from U.S. labor markets demonstrates that the deployment of robotics significantly reduces employment rates and wages, especially in regions with higher exposure to automation (Acemoglu & Restrepo, 2020).

Systematic reviews suggest that AI technologies are transforming human resource management practices, from recruitment to performance evaluation (Vrontis et al., 2021; Ekuma, 2023). While these advancements may reduce costs and improve efficiency, they also risk creating skill mismatches and necessitate adjustments in the labour market (Vrontis et al., 2021). The economic ramifications of AI and robotics also include increased productivity and potential wage growth in sectors that complement AI, but contribute to rising inequality, benefiting capital owners and skilled workers disproportionately (Murray, 2025; Yan & Grossman, 2023).

Although displacement risks are well documented, labour-market outcomes vary substantially across regions, suggesting that contextual features shape the capacity of economies to adapt. One set of explanations focuses on structural economic characteristics, including sectoral composition, innovation capacity, and institutional arrangements. A core concept in this literature is economic complexity, which measures the diversity and sophistication of a country's productive capabilities. Hidalgo and Hausmann (2009) argue that

economies with higher economic complexity possess broader knowledge networks and more flexible production structures, enabling them to diversify into new industries more easily. This adaptability is crucial when facing technology-driven disruptions because complex economies can reallocate workers into adjacent, high-skill activities that complement emerging technologies. Recent empirical work supports this view by linking higher complexity scores to greater innovation output, stronger long-term growth, and more robust responses to external shocks. *which work?*

Closely connected to economic complexity is the concept of economic resilience, which refers to an economy's ability to absorb, adapt to, and recover from shocks. In the context of automation, resilience encompasses not only productive diversification but also labour-market institutions, education and retraining systems, and regional governance capacity. Martin and Sunley (2015) describe resilience as a dynamic process shaped by historical pathways and structural conditions. Their framework suggests that regions with diversified industrial bases, effective retraining programs, and high levels of human capital demonstrate stronger adaptive capacity when confronted with structural change, including technological disruptions. Policy organisations, such as the OECD (2019), similarly argue that the impact of AI on employment depends significantly on countries' institutional frameworks, investment in skills, and the presence of active labour market policies.

This research aims to explore how robotics and AI are shaping the future of labor markets, and how the nation's economic complexity and resilience can reduce their negative effects. In this proposal, we will summarize our approach to answer these questions. First, we describe the data collection process, followed by the planned methodology for analyzing the data. Following these chapters, we will present our initial ideas about the research question and the results we aim to prove. Ultimately, we identify the beneficiaries of this research.

2. Description of data collection

For our research, we use data from Eurostat. It is the statistical office of the European Union, which cooperates with the statistical offices of the member states through the European Statistical System (ESS) to provide data that are comparable at the EU level. Within this framework is the Labour Force Survey, a quarterly representative sample (household) survey conducted by the national statistical offices. These surveys include individuals aged 15 and older, except those not living in private households, such as residents of nursing homes, prisoners, or those performing military service (Eurostat, 2025a; 2025b; 2025c; 2025d). A part of our variables originates from this database.

why this specification? what is the time frame?

We intend to use the activity rate and the average wage of a country, expressed in euros, as dependent variables. We aim to explain these with the share of firms in each country that use robots or artificial intelligence to perform various activities.

As a control variable, we plan to use a particular form of GDP and the Economic Complexity Index (ECI). This is an exception, because it comes from the Observatory of Economic Complexity (OEC) and not from the Eurostat (Observatory of Economic Complexity, 2025).

The most significant data loss occurs from the two main explanatory variables, as they are available for only a few time points. The artificial intelligence usage variable covers the years 2021, 2023, and 2024, while the robots' variable covers 2018, 2020, and 2022. When filtering for these years, we still have observations for many countries, which is shown in the figure below.

(alternative measures ??? eg. "AI" or "robots" Google Trends)

Figure 1: Countries planned to be included based on the preliminary filtering



source: own editing based on Eurostat data

3. Methodology

Since there is significant unobserved heterogeneity across observations (for example, countries' cultural characteristics or attitudes toward work), we use a fixed-effects model to filter out the time-invariant effect of each unit. The number of observations is considerably larger than the time dimension, so it is not advisable to include the observations as factor variables in a regression (for the purpose of filtering individual fixed effects). Instead, we apply within transformation.

we italics

In this case, for every observation i , we must calculate the time-averaged value of each variable, then subtract these from the observation for each time period t and estimate the resulting model using the ordinary least squares method. This way, the individual constants do not need to be estimated separately, yet they are still controlled for. Moreover, any correlation between the unit-specific constants and the error term disappears, indicating that time-invariant endogenous effects are handled (Wooldridge, 2010).

Finally, we also include time effects. However, if we were to include lagged values of the variables in such a static panel model, then at the observation level, the lagged dependent variable would be correlated with the average error term of the observation. This occurs because the averaging also includes the error term from one period earlier, and that earlier error term contains the explanatory variable from one period earlier, as it represents the regression estimation error relative to the actual y from the previous period. This problem arises even if the error term is otherwise not autocorrelated (Nickell, 1981). To avoid this, we applied an alternative solution in which the time effect is incorporated into the model through dummy time variables (Wooldridge, 2010).

This is essentially a more complex version of the difference-in-differences method, a popular research direction and methodology today, often used to conduct causal analysis (Angrist & Pischke, 2009). However, potential endogeneity may arise, and since the primary goal of our study is descriptive analysis, drawing causal conclusions is not our objective at this time.

Below is the model we plan to use, where α_i denotes the unit fixed effect and λ_t denotes the time fixed effect. The former captures the average value of each observation, while the latter captures the average value of a given year across all observations.

The above-mentioned variables will then be substituted into the places of the dependent (y) and explanatory (x) variables.

$$y_{it} = \beta_1 x_{it1} + \beta_2 x_{it2} + \dots + \beta_k x_{itk} + \alpha_i + \lambda_t + \varepsilon_{it} \quad (1)$$

4. Expected outcomes

Based on the existing literature on this topic, we anticipate a significant impact of AI and robotics on the European labour market. Since AI is a relatively new phenomenon, its impact is anticipated to be higher than that of robotics, which has developed alongside the industry since the Industrial Revolution.

5. Impact

The result can be helpful in several groups of stakeholders. First, policymakers can use our findings to understand better how the adoption of robots and artificial intelligence relates to labor market outcomes. This may support evidence-based decisions when designing technology-related regulations, labor market programs, or policies that aim to facilitate digital transformation while protecting workers.

Second, firms and employer organizations could use the results to assess how technological adoption in their sector compares with broader international patterns. Understanding these relationships can help them anticipate potential labor market adjustments or skill requirements associated with automation and AI.

Finally, the research may also be valuable for academic researchers and analysts studying technological change, labor economics, or European economic integration. By providing empirical evidence from multiple countries and years, the study can serve as a basis for further investigations or comparative analyses.

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